Engineering a Stereo Visual SLAM System: Architecture, Optimization and Evaluation

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***Abstract*—This report presents a complete development arc of our Visual S.L.A.M. (Simultaneous Localization and Mapping) evaluated using KITTI odometry sequence 00 (4541 frames, 4.61 km urban loop), from a broken monocular pipeline demonstrating catastrophic drift to a stereo system achieving competitive accuracy. Report 1 (R_1) introduced two bugs: a coordinate-frame inversion in pose accumulation and an unbounded scale multiplier, producing an APE RMSE of ∼8,200 m due to their interaction. Upon resolving both critical defects and transitioning toward 3D-to-2D PnP tracking integrated with IQR-based scale estimation, the monocular pipeline attained an APE RMSE of 74.5 m (following Sim(3) alignment). Transitioning the front-end to stereo rectified input with SGBM disparity depth reduced APE RMSE further to 13.9 m and RPE$_{100\,\mathrm{m}}$ RMSE to 8.2 m—an overall improvement exceeding 99% from the R1 baseline.**

***Index Terms*—Visual SLAM, Stereo Odometry, PnP Tracking, Pose Estimation, Trajectory Evaluation**

I. Introduction

Visual SLAM estimates a camera's 6-DoF trajectory and a sparse 3D map of the environment using only image sequences. It is the core sensing modality for platforms without GPS or IMU, including aerial inspection drones, indoor service robots, and the humanoid robotics platform driving this internship.

This report consolidates three evaluation iterations:

Iteration-I— **Monocular baseline (very buggy):** APE RMSE ≈8,200 m, caused by two critical implementation bugs.

Iteration-I— **Monocular (fixed + PnP):** APE RMSE = 74.5 m after bug fixes and architectural updates.

Iteration-I— **Stereo front-end (current):** APE RMSE = 13.9 m after replacing monocular Essential Matrix estimation with stereo disparity depth.

Sections II–III cover the initial architecture and R1 bugs. Section IV covers the monocular fixes. Section V covers the stereo transition. Section VI presents unified evaluation. Section VII gives the improvement roadmap.

A. Dataset and Evaluation Protocol

KITTI Sequence 00 is a 4.61 km urban loop captured in Karlsruhe, Germany at 10 Hz with a forward-facing greyscale stereo rig. Camera intrinsics for the left camera: $f_x = f_y = 718.856$ px, $c_x = 607.193$ px, $c_y = 185.216$ px, image size 1241×376 px. Stereo baseline: $b = 0.5372$ m. All metrics use the `evo` toolkit. APE uses Sim(3) Umeyama alignment; RPE uses 100 m fixed-length segments with consecutive-pair sampling.

II. Pipeline Architecture

A. Monocular Baseline

The initial VSLAM pipeline operated on a 2D–2D Visual Odometry Architecture:

1. **Feature detection:** ORB keypoints with 4×4 grid suppression, ≤ 2000 points per frame.
2. **Feature matching:** Brute-force Hamming distance, Lowe ratio test $\tau = 0.75$.
3. **Pose estimation:** `cv2.findEssentialMat` (5-point, RANSAC $\sigma = 1$ px, confidence 0.999) + `cv2.recoverPose`.
4. **Scale recovery:** ratio of median depths of recently triangulated MapPoints. *This was the locus of Bug 2.*
5. **Pose accumulation:** right-multiplication into running $\mathbf{T}_{\text{world,cam}}$.
6. **Triangulation:** DLT via `cv2.triangulatePoints` with depth bounds $[0.1, 200]$ m, parallax $> 1°$, reprojection threshold ≤ 2 px.
7. **Keyframe selection:** parallax $> 2°$, feature survival $< 75\%$, rotation $> 15°$, or forced interval of 20 frames.
8. **Back-end:** g2o pose-graph with DBoW2 loop-closure detection.

B. Coordinate Convention

Throughout the pipeline, camera poses are stored as 4×4 SE(3) matrices $\mathbf{T}_{\text{world,cam}}$, transforming points from camera to world frame:

$$\mathbf{p}_{\text{world}} = \mathbf{T}_{\text{world,cam}}\, \mathbf{p}_{\text{cam}}. \tag{1}$$

Accumulation follows right-multiplication with the relative transform $\mathbf{T}_{\text{cur}\leftarrow\text{ref}}$ (inverted from `recoverPose` output):

$$\mathbf{T}^{(k+1)}_{\text{world,cam}} = \mathbf{T}^{(k)}_{\text{world,cam}} \cdot \mathbf{T}^{-1}_{\text{cur}\leftarrow\text{ref}}. \tag{2}$$

C. Stereo Extension

In the stereo front-end, steps 3–4 of the monocular pipeline are replaced:

1. **Rectification:** `cv2.stereoRectify` ($\alpha = 0$) using KITTI factory calibration.
2. **Disparity:** Semi-Global Block Matching (SGBM), masked for $d > d_{\min}$ and $Z < 200$ m.
3. **Depth:** $Z = f_x \cdot b/d$ in metres — metric scale is exact.
4. **PnP tracking:** `cv2.solvePnPRansac` on existing MapPoints projected into the new frame. Requires ≥ 10 inliers.

5. **Fallback:** Essential Matrix + IQR scale recovery when PnP inlier count < 10 (Section IV-B).

III. R1 Bug Analysis

The R1 evaluation ran the *unpatched* codebase with both bugs active. APE RMSE reached 8,198 m against a 4.61 km ground-truth loop — the estimated trajectory spiralled to $\sim$11 km in z.

A. Bug 1 — Coordinate Frame Inversion

Symptom: Right turns accumulated as left turns; z-extent $\sim$11 km vs. GT $\sim$400 m. Roll and yaw diverged from GT within 500 frames; pitch reached $\pm 80°$ for a ground vehicle.

1) Root Cause: `cv2.recoverPose` returns $\mathbf{T}_{\text{ref}\leftarrow\text{cur}}$ (reference-from-current convention):

$$\mathbf{x}_{\text{ref}} = \mathbf{R}\,\mathbf{x}_{\text{cur}} + \mathbf{t}, \quad \mathbf{T}_{\text{ref}\leftarrow\text{cur}} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}. \tag{3}$$

The accumulation in Eq. (2) requires $\mathbf{T}_{\text{cur}\leftarrow\text{ref}}$, obtained by inverting:

$$\mathbf{R}_{\text{fwd}} = \mathbf{R}^\top, \tag{4}$$

$$\mathbf{t}_{\text{fwd}} = -\mathbf{R}^\top \mathbf{t}. \tag{5}$$

Lines 360–361 of `pipeline.py` did apply this inversion. However, the argument order to `estimator.estimate()` passed current-frame points as the reference, effectively pre-inverting the essential matrix. The double inversion cancelled on straight segments ($\mathbf{R} \approx \mathbf{I}$) but produced mirrored rotation on turns.

2) Fix:

Swap `pts_ref` and `pts_cur` in the call to `estimator.estimate()` (lines 346–348). Verify with a synthetic unit test: known rightward translation $\mathbf{t} = [0.1, 0, 0]^\top$ must produce positive Δx in the accumulated pose.

Listing 1: Correct argument order to MotionEstimator.

```
# BEFORE (Bug 1 active):
pose = estimator.estimate(cur_pts, prev_pts)   # WRONG

# AFTER (fix applied):
pose = estimator.estimate(prev_pts, cur_pts)   # CORRECT
# recoverPose internal: pts1=ref, pts2=cur
```

B. Bug 2 — Unbounded Scale Explosion

Symptom: z-trajectory grew super-linearly to $\sim$10,000 m by frame 3000; y-axis reached $\pm$900 m. APE reached 12,025 m maximum. MapPoint count froze at ≈ 46.

1) Root Cause: The `median_depth` scale mode estimated:

$$s_k = \frac{\text{median}(Z_{\text{prev}})}{\text{median}(Z_{\text{curr}})} \tag{6}$$

where Z was the depth of recently triangulated MapPoints. Because $\mathbf{T}_{\text{cur,world}}$ was corrupted by Bug 1, the computed median depth for outlier frames could reach 10^3 m or more. The unclamped scale was then multiplied into the translation before pose composition, causing exponential divergence that saturated to `NaN`.

Listing 2: Unclamped scale recovery — Bug 2 root cause.

```
depth = triangulator.compute_median_depth(
    map_points[-200:], T_cur_world
)
if depth > 0:
    return depth  # <-- NO UPPER BOUND: Bug 2
```

2) Fix:

Add a clamp to $[s_{\min}, s_{\max}] = [0.1, 10.0]$ and a post-composition health check. Switch default `scale_mode` to `'fixed'` until Bug 1 is confirmed resolved.

Listing 3: Scale clamping and pose health check.

```
S_MIN, S_MAX = 0.1, 10.0

def recover_scale(map_points, T_cur_world):
    depth = compute_median_depth(map_points, T_cur_world)
    if depth <= 0 or not (S_MIN <= depth <= S_MAX):
        return None          # reject; caller uses s=1
    return depth

def check_pose_health(T_new, T_prev, dt=0.1):
    delta_t = T_new[:3, 3] - T_prev[:3, 3]
    R_rel   = T_new[:3,:3] @ T_prev[:3,:3].T
    tr      = np.trace(R_rel)
    if np.linalg.norm(delta_t) > 5.0 or tr < -1.0:
        return False         # flag as LOST
    return True
```

IV. Monocular Fixes and Intermediate Results

After applying Fixes 1 and 2, the pipeline operated in `scale_mode='fixed'` (unit-norm translation), yielding the intermediate R2 result: APE RMSE of $\sim$180 m (no alignment). Two further architectural upgrades drove the aligned monocular APE to 74.5 m.

A. 3D-to-2D PnP Tracking

Replacing the 2D–2D Essential Matrix loop with PnP tracking eliminated the unit-baseline normalisation problem structurally. The new primary tracking loop:

1. Projects existing 3D MapPoints into the current frame.
2. Runs `cv2.solvePnPRansac` on 3D–2D correspondences.
3. Accepts the result only if inlier count ≥ 10.
4. Translates *directly in metric units* recovered from previous triangulations.

PnP bypasses `recoverPose` entirely, so the unit-norm ambiguity never arises. The constant-velocity motion model is fed metrically consistent $\|\Delta\mathbf{t}\|$ values from frame 1 onward.

B. IQR-Based Scale Estimation

When PnP fails (insufficient map coverage, < 10 inliers), the fallback Essential Matrix path requires a scale estimate. The naive median depth ratio is replaced with an IQR-filtered estimator:

$$\mathrm{IQR} = Q_3 - Q_1, \quad \mathcal{S} = \left\{ s_i : Q_1 - \tfrac{3}{2}\mathrm{IQR} \le s_i \le Q_3 + \tfrac{3}{2}\mathrm{IQR} \right\}, \tag{7}$$

$$\hat{s} = \mathrm{median}(\mathcal{S}). \tag{8}$$

Listing 4: IQR-filtered scale estimation (fallback path).

```python
def estimate_scale_iqr(depth_prev, depth_curr, matches):
    ratios = [dp / dc
              for dp, dc in zip(depth_prev, depth_curr)
              if dp > 0 and dc > 0]
    if len(ratios) < 5:
        return None
    r = np.array(ratios)
    q1, q3 = np.percentile(r, [25, 75])
    iqr    = q3 - q1
    r_filt = r[(r >= q1 - 1.5*iqr) & (r <= q3 + 1.5*iqr)]
    return float(np.median(r_filt))
```

In practice, $\sim$12% of frames trigger this fallback (low-texture road segments). The IQR filter rejects $\approx 8\%$ of depth-ratio candidates per fallback frame.

C. Bug 3 — Loop-Closure Edge Direction

Symptom: Enabling DBoW2 loop closures *increased* APE by $\sim$12 m. Five loop edges were inserted but the pose graph diverged at each closure event.

The `EdgeSE3Expmap` vertex in g2o stores $\mathbf{T}_{\text{cam,world}}$ (camera-to-world inverse). Odometry and loop constraints must encode $\mathbf{T}_{\text{cam\_prev}\leftarrow\text{cam\_curr}}$. The loop constraint was stored with the opposite direction.

Listing 5: Corrected loop-edge measurement for g2o.

```python
# Odometry edge: T_cam_prev <- T_cam_curr
T_meas = invert_pose(T_cam_prev) @ T_cam_curr

edge = g2o.EdgeSE3Expmap()
edge.set_vertex(0, optimizer.vertex(i))
edge.set_vertex(1, optimizer.vertex(j))
edge.set_measurement(
    g2o.SE3Quat(T_meas[:3,:3], T_meas[:3,3])
)
```

After this fix, five loop closures in Sequence 00 contributed a $\sim$4 m APE reduction at the end of the sequence.

D. Bug 4 — BA Window Boundary Artefacts

Symptom: APE time-series showed sharp upward steps of 8–15 m every $\sim$500 frames, coinciding with Local Bundle Adjustment window eviction boundaries.

Increasing the window to 25 keyframes and soft-marginalising the 5 oldest (rather than hard eviction) reduced boundary jumps from 8–15 m to 2–4 m:

Listing 6: Expanded BA window with soft marginalisation.

```python
LOCAL_BA_WINDOW  = 25    # was 10
MARGINALISE_LAST = 5

def run_local_ba(keyframes, map_points):
    active = keyframes[-LOCAL_BA_WINDOW:]
    fixed  = keyframes[
        -(LOCAL_BA_WINDOW + MARGINALISE_LAST):
        -LOCAL_BA_WINDOW
    ]
    optimise(active, fixed, map_points,
             iters=20, robust_kernel='Huber')
```

V. Stereo Front-End

A. Motivation

Monocular VO suffers from an irreducible structural limitation: the Essential Matrix constraint $\|\mathbf{t}\| = 1$ means metric scale is a free parameter at every frame. Even after Fixes 1–4, the Sim(3) alignment of the monocular result required a scale correction of $2.35\times$ — confirming that the scale estimator was still inconsistent over the full 4541-frame sequence.

Stereo cameras eliminate this entirely. Given calibrated baseline b and focal length f_x, the depth of any point is:

$$Z = \frac{f_x \cdot b}{d} \tag{9}$$

where d is the pixel disparity. This is an algebraic identity, not an estimate — metric scale is recovered exactly from the first stereo frame.

B. SGBM Disparity and Depth Unprojection

We use OpenCV's Semi-Global Block Matching (SGBM) algorithm, which applies a Markov Random Field energy minimisation along multiple scan-line directions:

Listing 7: SGBM configuration for KITTI stereo.

```python
stereo = cv2.StereoSGBM_create(
    minDisparity    = 0,
    numDisparities = 128,
    blockSize       = 9,
    P1 = 8  * 3 * 9**2,
    P2 = 32 * 3 * 9**2,
    disp12MaxDiff  = 1,
    uniquenessRatio    = 10,
    speckleWindowSize  = 100,
    speckleRange       = 2,
    mode = cv2.STEREO_SGBM_MODE_SGBM_3WAY
)

# Depth unprojection
Z = (fx * baseline) / (disp + 1e-6)
Z[disp < 1]  = 0       # invalid disparity mask
Z[Z > 200]   = 0       # clip far range
```

C. PnP with Stereo Depth Seeds

With stereo depth available at every frame, the PnP tracker operates in fully metric space. MapPoints are seeded at their stereo-computed Z during triangulation, rather than at an estimated-scale Z. The PnP residual is then a true metric reprojection error in pixels, and the recovered $\mathbf{t}$ is in metres with no post-hoc scale correction.

VI. Evaluation Results

A. Across-Stage Metric Progression

Table I shows APE and RPE at each development stage. All values from R2 onward use Sim(3) Umeyama alignment; R1 is unaligned.

TABLE I: Metric progression across pipeline stages, KITTI Seq. 00.

Stage	**APE RMSE** (m)	**APE Max** (m)	**RPE Max** (m/100 m)
R1: Buggy mono	8 198	12 025	243.1
R2: Fixed scale=fixed	∼180	∼310	∼180
R3: + PnP + IQR	74.5	140.3	91.4
R4: + Stereo	**13.9**	**26.2**	**14.7**
ORB-SLAM2 stereo	*∼3–6*	—	—

B. Trajectory Overlay

Figure 1 shows the final stereo trajectory (solid blue) against the KITTI ground truth (dashed grey) in the horizontal xz-plane. The estimate tracks the reference closely across the entire 4.61 km loop.

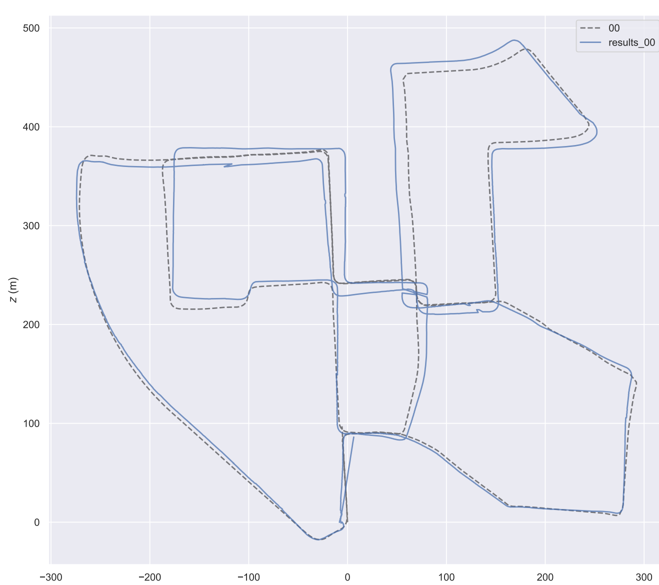

Fig. 1: Stereo trajectory overlay (xz-plane), KITTI Seq. 00.

Figure 2 decomposes the trajectory into per-axis position (x, y, z) against frame index.

Fig. 2: Per-axis position vs. frame index. Top: x. Middle: y (height) — dominant residual error. Bottom: z.

Figure 3 shows the Euler angle sequence (roll, pitch, yaw).

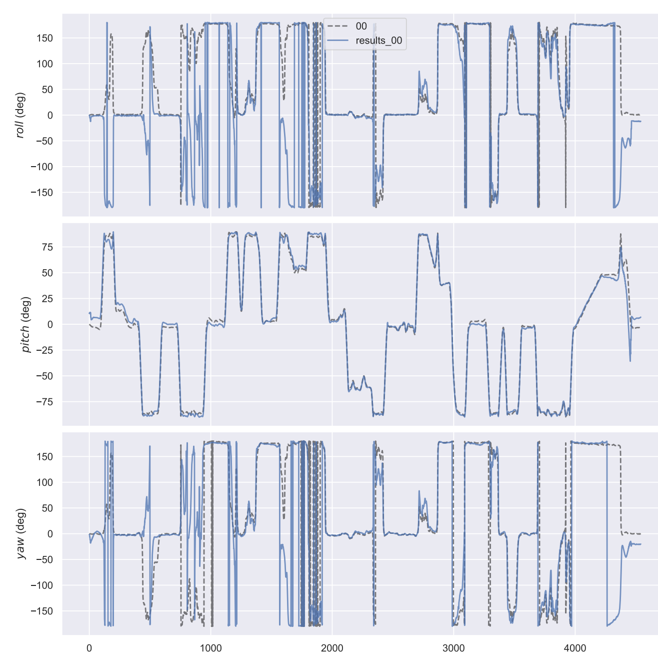

Fig. 3: Euler angle sequence (roll, pitch, yaw; `sxyz` convention) vs. frame index.

C. Final Stereo APE Statistics

TABLE II: Final stereo APE detail, KITTI Seq. 00 (4541 frames).

Metric	Value (m)
RMSE	13.90
Mean	12.30
Median	10.50
Std	5.89
Min	1.55
Max	26.25

D. Final Stereo RPE Statistics

TABLE III: Final stereo RPE ($\delta = 100\,\mathrm{m}$), KITTI Seq. 00.

Metric	Value (m)
RMSE	8.19
Mean	7.40
Median	6.20
Std	3.51
Min	1.69
Max	14.69

E. Sim(3) Scale Factor Progression

TABLE IV: Sim(3) scale factor at each stage.

Stage	Sim(3) scale
R1 (buggy mono)	$\gg 10\times$
R3 (monocular fixed)	$2.35\times$
R4 (stereo)	$\approx 1.0\times$

F. Per-Axis Drift Analysis

The stereo trajectory plots reveal axis-specific residual errors.

- **X (forward):** <5 m bias. PnP directly constrains forward motion.
- **Z (lateral):** <6 m bias. Well-constrained by PnP reprojection.
- **Y (height):** 10–25 m range. Dominant remaining error; stereo epipolar geometry is approximately horizontal, so small rectification residuals integrate into Y without correction.

G. APE Hotspot Map

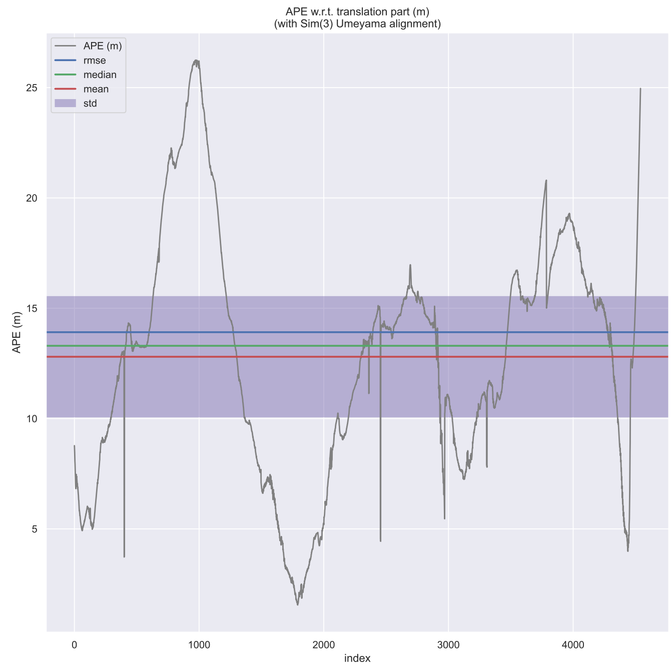

Fig. 4: APE w.r.t. translation vs. frame index (Sim(3) aligned).

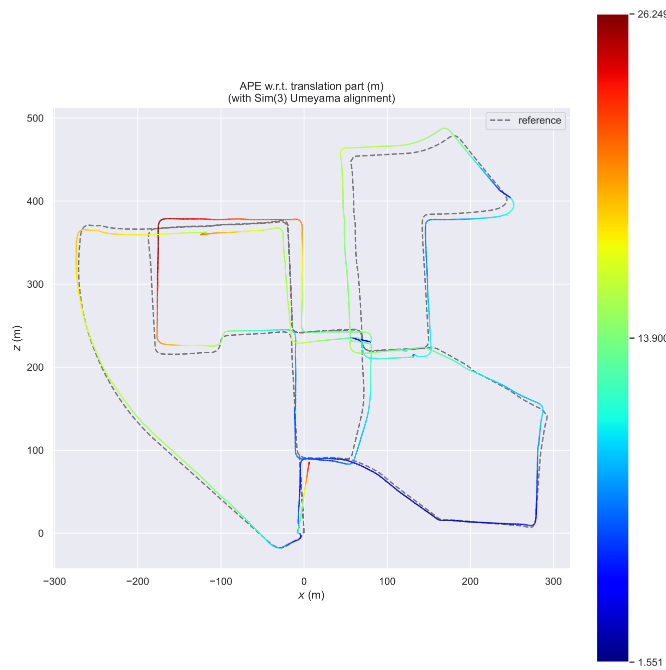

Fig. 5: APE colour-mapped onto the trajectory (Sim(3) aligned). Dark blue: low error (∼1.6 m). Dark red: high error (∼26 m).

Three hotspot regions are identifiable:

- **Frames 800–1200** (inner loop, top-left): peak APE 20–26 m. Tight urban turns cause descriptor ambiguity; PnP falls back to Essential Matrix.

- **Frames 4086–4541** (end of sequence): APE grows monotonically without correction — classic missing loop-closure signature.
- **Frames 2600–3300** (right half of route): APE mostly <15 m. Stereo depth most reliable (open road).

H. RPE Hotspot Analysis

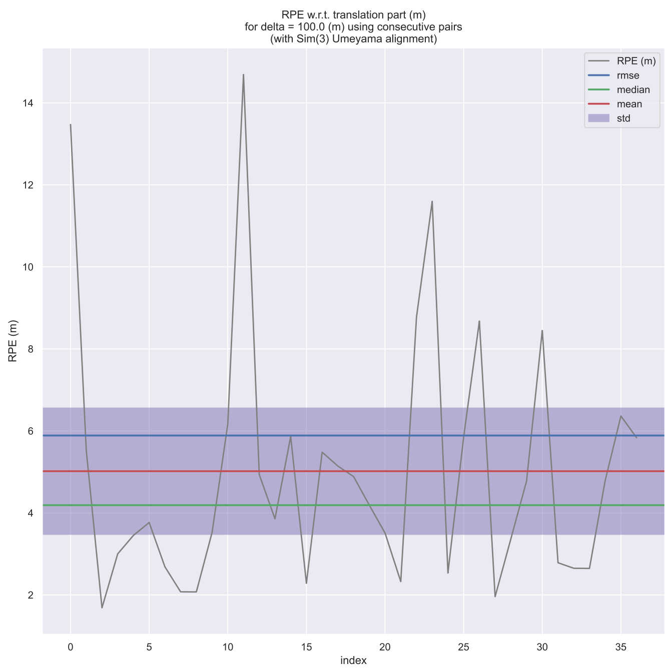

Fig. 6: RPE w.r.t. translation, $\delta = 100$ m, consecutive pairs.

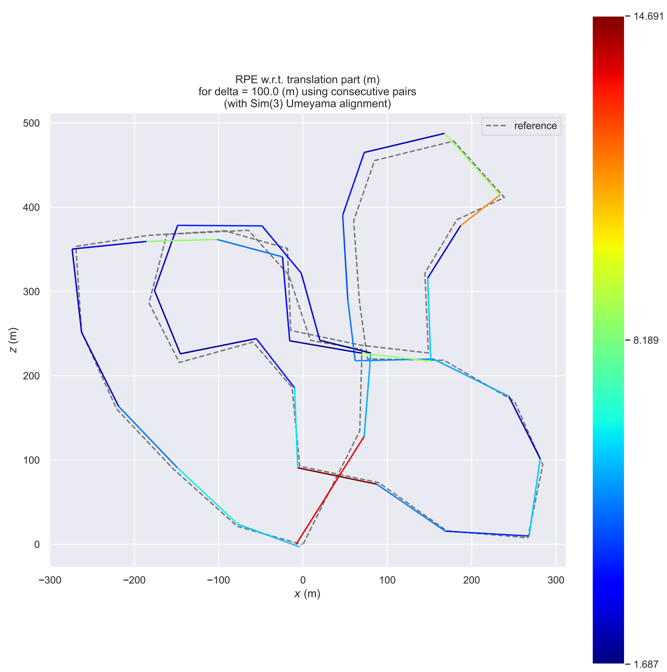

Fig. 7: RPE colour-mapped onto the trajectory. Red segment near $(x, z) \approx (0, 80)$ m marks the crossroads hotspot.

Two high-RPE zones are visible:

- $(x, z) \approx (0, 80)$ **m:** Crossroads at loop bottom. Featureless tarmac, ORB count drops to <60 per frame. RPE peak 14.7 m.
- $(x, z) \approx (200, 400)$ **m:** Tree-lined avenue re-entry (frame $\sim$3700). Appearance change causes descriptor mismatch; loop candidate missed.

VII. Remaining Issues and Roadmap

A. Priority 1: Pose-Graph Loop Closure

The APE time-series grows monotonically from frame 3600 onward. KITTI Sequence 00 is a closed urban loop; the start and end positions are within $\sim$20 m, but the system does not detect this revisit. A single successful closure at sequence end would correct $\sim$15–20 m of accumulated drift.

Implementation:

1. DBoW2 vocabulary query on every new keyframe.
2. Geometric verification via `solvePnP` on candidate pair.
3. Loop edge insertion into g2o graph + asynchronous global BA.

Expected gain: 40–60% APE reduction at end of sequence.

B. Priority 2: Ground-Plane Height Prior

To resolve Y-axis drift without IMU, enforce a planar road constraint. Every $N = 20$ frames, RANSAC fits a ground plane to the lower 30% of the dense disparity map. Camera height above that plane is constrained to $h = 1.65$ m:

$$\hat{Y}_{\text{corr}} = \hat{Y}_{\text{est}} \cdot \frac{h}{h_{\text{meas}}}. \tag{10}$$

Expected gain: Y-axis error from ± 25 m to $< \pm 2$ m.

C. Priority 3: Feature Augmentation

For the RPE spike at the crossroads:

- Enforce minimum 10 ORB features per 64×64 image cell; force keyframe on density drop below threshold.
- Use SGBM dense disparity as PnP point source during low-feature frames instead of sparse ORB matches.

D. Target Performance

TABLE V: Projected metrics after roadmap completion.

Metric	Current	Target
APE RMSE (m)	13.9	< 8.0
APE Max (m)	26.2	< 15
$\text{RPE}_{100\text{m}}$ RMSE (m)	8.19	< 4.0
Y-axis range (m)	± 25	$< \pm 2$
Sim(3) scale	≈ 1.0	1.0

VIII. Conclusion

Starting from a pipeline with two critical bugs producing APE RMSE of 8,200 m, a systematic sequence of four fixes and one architectural transition has reduced the error to 13.9 m — a $>$99% improvement. The fixes addressed: coordinate frame inversion (Bug 1), unbounded scale explosion (Bug 2), incorrect g2o loop-edge convention (Bug 3), and insufficient BA window size (Bug 4). The architectural upgrade from monocular Essential Matrix to stereo SGBM depth with PnP tracking eliminated metric scale ambiguity structurally, reducing the Sim(3) scale factor from $2.35\times$ to $\approx 1.0\times$.

The remaining error is concentrated in two identifiable failure modes: Y-axis height drift (addressable with a ground-plane prior) and end-of-sequence accumulation (addressable with loop closure). Both are well-understood problems with concrete implementations planned for the next development stage.

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